

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the limit $\lim_{x \rightarrow 4} (4x - 9) = 7$.
- a) Suppose that someone challenged you to show that the y -values of $f(x) = 4x - 9$ can get within 4 units from 7. Determine a value of δ that would make this possible.

We will start by investigating the behavior of the y -values by looking at the interval $|4x - 9 - 7| < 4$.

We see that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4|.$$

So, if $|x - 4| < 1$ then we could say that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4| < 4(1) = 4$$

and therefore $|4x - 9 - 7| < 4$.

Therefore we need $\delta = 1$. We have showed that if we pick an x -value in $|x - 4| < 1$ (meaning x is in $(3, 5)$) then $|f(x) - 7| < 4$ (meaning $f(x)$ is in $(3, 11)$).

- b) Suppose that someone challenged you to show that the y -values of $f(x) = 4x - 9$ can get within 2 units from 7. Determine a value of δ that would make this possible.

We will start by investigating the behavior of the y -values by looking at the interval $|4x - 9 - 7| < 2$.

We see that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4|.$$

So, if $|x - 4| < \frac{1}{2}$ then we could say that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4| < 4\left(\frac{1}{2}\right) = 2$$

and therefore $|4x - 9 - 7| < 2$.

Therefore we need $\delta = \frac{1}{2}$. We have showed that if we pick an x -value in $|x - 4| < \frac{1}{2}$ (meaning x is in $(3.5, 4.5)$) then $|f(x) - 7| < 2$ (meaning $f(x)$ is in $(5, 9)$).

- c) Suppose that someone challenged you to show that the y -values of $f(x) = 4x - 9$ can get within $\frac{1}{2}$ units from 5. Determine a value of δ that would make this possible.

We will start by investigating the behavior of the y -values by looking at the interval $|4x - 9 - 7| < \frac{1}{2}$.

We see that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4|.$$

So, if $|x - 4| < \frac{1}{8}$ then we could say that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4| < 4\left(\frac{1}{8}\right) = \frac{1}{2}$$

and therefore $|4x - 9 - 7| < \frac{1}{2}$.

Therefore we need $\delta = \frac{1}{8}$. We have showed that if we pick an x -value in $|x - 4| < \frac{1}{8}$ (meaning x is in $(3.875, 4.125)$) then $|f(x) - 7| < \frac{1}{2}$ (meaning $f(x)$ is in $(6.5, 7.5)$).

- d) Suppose that someone challenged you to show that the y -values of $f(x) = 4x - 9$ can get within ε units from 5. Determine a value of δ that would make this possible.

We will start by investigating the behavior of the y -values by looking at the interval $|4x - 9 - 7| < \varepsilon$.

We see that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4|.$$

So, if $|x - 4| < \frac{\varepsilon}{4}$ then we could say that

$$|4x - 9 - 7| = |4x - 16| = |4(x - 4)| = 4|x - 4| < 4\left(\frac{\varepsilon}{4}\right) = \varepsilon$$

and therefore $|4x - 9 - 7| < \varepsilon$.

Therefore we need $\delta = \frac{\varepsilon}{4}$. We have showed that if we pick an x -value in $|x - 4| < \frac{\varepsilon}{4}$ (meaning x is in $\left(4 - \frac{\varepsilon}{4}, 4 + \frac{\varepsilon}{4}\right)$) then $|f(x) - 7| < \varepsilon$ (meaning $f(x)$ is in $(7 - \varepsilon, 7 + \varepsilon)$).

e) Use the epsilon-delta definition of a limit to prove that $\lim_{x \rightarrow 4} (4x - 9) = 7$.

Scratch work: We want to pick a delta such that: “if $0 < |x - 4| < \delta$, then $|4x - 9 - (7)| < \varepsilon$ ”. So, we have that $0 < |x - 4| < \delta$. We note that:

$$|4x - 9 - (7)| = |4x - 16| = |4(x - 4)| = 4|x - 4| < 4 \cdot \delta$$

Notice that if we take $\delta = \frac{\varepsilon}{4}$, then $4\delta = \varepsilon$, and we will have that the if-then statement holds.

Solution: Suppose ε is some positive real number. Let $\delta = \frac{\varepsilon}{4}$, and suppose that $0 < |x - 4| < \delta$. Then,

$$|4x - 9 - (7)| = |4x - 16| = |4(x - 4)| = 4|x - 4| < 4 \cdot \delta = \varepsilon$$

Therefore, we can say that $\lim_{x \rightarrow 4} (4x - 9) = 7$.

2. Determine the limit of $\lim_{x \rightarrow 1} \left(\frac{5-x}{2} \right)$ and prove the limit is correct.

We guess that $\lim_{x \rightarrow 1} \left(\frac{5-x}{2} \right) = 2$.

Scratch work: We want to pick a delta such that: “if $0 < |x - 1| < \delta$, then $\left| \frac{5-x}{2} - 2 \right| < \varepsilon$ ”. So, we have

that $0 < |x - 1| < \delta$. We note that: $\left| \frac{5-x}{2} - 2 \right| = \left| \frac{1-x}{2} \right| = \left| \frac{1}{2}(1-x) \right| = \left| -\frac{1}{2}(x-1) \right| = \frac{1}{2}|x-1| < \frac{1}{2} \cdot \delta$

Notice that if we take $\delta = 2\varepsilon$, then $\frac{1}{2}\delta = \varepsilon$, and we will have that the if-then statement holds.

Solution: Suppose ε is some positive real number. Let $\delta = 2\varepsilon$, and suppose that $0 < |x - 1| < \delta$. Then,

$$\left| \frac{5-x}{2} - 2 \right| = \left| \frac{1-x}{2} \right| = \left| \frac{1}{2}(1-x) \right| = \left| -\frac{1}{2}(x-1) \right| = \frac{1}{2}|x-1| < \frac{1}{2} \cdot \delta = \varepsilon$$

Therefore, we can say that $\lim_{x \rightarrow 1} \left(\frac{5-x}{2} \right) = 2$.

Discussion: Talk about that the definition allows us to show that the y-values of our function become as close as we could ever want to the value of the limit (which implies that they approach the distance of 0). Since we are using variables we are arguing for *every possible epsilon* value simultaneously!